ModelIt!

Teacher's Guide

Can We Feed 10 Billion People?

Modeling Agricultural Systems, Land Use, and Global Food Security

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-1

Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to calculate food system carrying capacity under different scenarios, quantifying the relationships between population growth, resource constraints, and food production potential.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — Ecosystems have carrying capacities that result from biotic and abiotic factors, and the carrying capacity of agricultural ecosystems depends on maintaining the soil biology, water cycles, and biodiversity that sustain productivity.
Crosscutting Concept	**Systems and System Models** — Students use systems thinking to understand how food production, resource consumption, waste, and environmental degradation form interconnected feedback loops that determine whether food systems are sustainable or self-undermining.

CAST Testing Alignment

MULTIPLE CHOICE:

A region increases fertilizer application by 40% to boost crop yields. In the first 5 years, yields increase by 15%. But after 20 years, yields are lower than before the fertilizer increase. Based on your food system model, which explanation is most accurate? A) The fertilizer was the wrong type B) Soil Health Index declined due to biological disruption, canceling out chemical gains C) The crops developed immunity to fertilizer D) Rainfall decreased independently

CONSTRUCTED RESPONSE:

Using your food system model, explain why reducing food waste by 50% could be more effective than increasing agricultural land by 20% for feeding a growing population. Reference at least three components and the concept of carrying capacity in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The global food system is one of humanity\'s greatest achievements and greatest vulnerabilities. Modern agriculture feeds nearly 8 billion people — more than any point in history — but does so by depleting the soil, water, and biodiversity that future food production depends on. We are essentially borrowing from the future to feed the present, and the bill is coming due as population grows toward 10 billion while agricultural resources degrade.

The Green Revolution: Triumph and Trade-offs

The Green Revolution of the 1960s-1980s dramatically increased global food production through high-yield crop varieties, synthetic fertilizers, pesticides, and irrigation. Global grain production tripled between 1950 and 2000, preventing predicted mass famines. However, this success came with hidden costs: dependence on fossil fuel-based inputs, groundwater depletion from massive irrigation, soil degradation from monoculture and tillage, biodiversity loss from pesticides and habitat conversion, and dead zones in coastal waters from fertilizer runoff. The Green Revolution showed that technology can dramatically increase production, but it did not solve the underlying challenge of sustainability — it delayed it while making the eventual reckoning larger.

Soil: The Foundation Under Threat

Healthy agricultural soil is a living ecosystem containing billions of bacteria, fungi, insects, and worms per gram that cycle nutrients, retain water, suppress disease, and build soil structure. Intensive monoculture, chemical fertilizers, and tillage disrupt this biology, leading to soil that is chemically fed but biologically dead. The consequences are severe: the United Nations estimates that one-third of global topsoil is already degraded, and at current rates of loss, the world has about 60 harvests left in many regions. Regenerative agriculture — cover crops, reduced tillage, composting, crop rotation, and integrated livestock — can reverse soil degradation, but the transition takes 3-7 years and requires support during the yield dip that often occurs initially.

The Water Crisis in Agriculture

Agriculture consumes 70% of global freshwater withdrawals, primarily for irrigation. Many of the world\'s most productive agricultural regions — the US Great Plains, India\'s Punjab, China\'s North China Plain — depend on groundwater aquifers that are being depleted far faster than they recharge. The Ogallala Aquifer under the US Great Plains has lost 30% of its volume since 1960 and continues declining. When these aquifers are exhausted, the agricultural production they support will collapse — and unlike soil, depleted aquifers cannot be regenerated on any human timescale. More efficient irrigation (drip systems, soil moisture monitoring) can reduce water use by 30-50%, but ultimately, water-intensive crops in arid regions represent an unsustainable model.

Food Waste: The Hidden Opportunity

Approximately one-third of all food produced globally — 1.3 billion tons worth $1 trillion per year — is lost or wasted. In developing countries, losses occur primarily after harvest due to inadequate storage, transport, and processing infrastructure. In developed countries, waste occurs at the retail and consumer level due to cosmetic standards, oversized portions, confusion about expiration dates, and cultural devaluation of food. Reducing food waste by 50% would effectively increase the global food supply by 16% without requiring a single additional acre of farmland or gallon of water. This is the most cost-effective, environmentally beneficial, and technically feasible intervention in the entire food system — yet it receives a fraction of the investment directed at increasing production.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Global Population, Agricultural Land Area, Crop Yield Per Hectare, Soil Health Index, Water Availability, Food Waste and Loss, and Biodiversity in Agricultural Landscapes as key components of the food system.

E **Establish Relationships**

Students determine that population growth drives food demand, that intensive farming degrades soil and water over time, that food waste represents the largest inefficiency, and that biodiversity provides essential ecosystem services for sustainable production.

V **Visualize & Model**

Students build a computational model showing how food system components interact over time, revealing the tension between short-term production maximization and long-term sustainability.

E **Evaluate Outcomes**

Students run industrial intensification, population pressure, and sustainable intensification scenarios to test predictions about food system capacity and the effectiveness of different intervention strategies.

R **Revise & Extend**

Students add dietary composition, climate change impacts, or agricultural technology to explore how additional factors affect food system carrying capacity and sustainability.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"By 2050, there will be 10 billion people on this planet. Can we feed them all without destroying the ecosystems we depend on? That\'s what we\'re modeling today."*

Watch For

Show cover image. Quick poll: How many people think we can feed 10 billion? How many think we can\'t? What would need to change?

Slide 2: Learning Objectives

3 min

Say: *"Today we\'re modeling the global food system — not just farming, but the entire chain from soil to plate, including what gets wasted along the way."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key insight: carrying capacity isn\'t just about maximum production — it\'s about sustainable production over time.

Slide 3: Big Question

3 min

Say: *"We already produce enough calories to feed 10 billion people. So why are 800 million hungry? And what happens when demand grows while soil and water decline?"*

Watch For

Quick-write: Students write their initial analysis of the food system challenge. Is it a production problem, a distribution problem, or both?

Slide 4: LEVER Framework

2 min

Say: *"We\'ll build a model that reveals the hidden dynamics of the food system — including why producing more food can sometimes make the problem worse in the long run."*

Watch For

Preview each LEVER step. Emphasize that this model will challenge assumptions about whether the food crisis is simply about growing more food.

Slide 5: Activity 1: Components

8 min

Say: *"Sort these into external drivers and internal system responses. Think about what inputs the food system receives versus what happens inside the system."*

Watch For

Guide sorting: External = Global Population (demographic trend), Agricultural Land Area (policy decision). Internal = the rest. Discuss why Crop Yield is internal even though it feels like an output.

Slide 6: Activity 2: Connections

8 min

Say: *"How does increasing chemical inputs boost short-term yields while degrading long-term soil health? Trace the feedback loop between soil, yield, and the pressure to add more chemicals."*

Watch For

Students draw arrows. Key discovery: the vicious cycle of soil degradation — declining soil health requires more chemical inputs, which further degrade biology, requiring even more inputs. Compare to the sustainable cycle of regenerative practices.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios and pay attention to what happens in year 5 versus year 30. The short-term and long-term results may be very different."*

Watch For

Students run scenarios, compare predictions to observations. Key discussion: Why does industrial intensification look good in year 5 but catastrophic in year 30? What changed?

Slide 8: Discoveries

5 min

Say: *"The biggest surprise from our model: the most impactful intervention isn\'t producing more food — it\'s wasting less. Let\'s discuss why."*

Watch For

Class discussion. Compare initial predictions to model evidence. Challenge: If we already produce enough food for 10 billion, why do we focus on production instead of waste and distribution?

Slide 9: STEM Challenge

12 min

Say: *"Your city is doubling in population in a semi-arid region with a declining aquifer. Design a food system that\'s still working in 50 years. Use your model data."*

Watch For

Groups design food system plans addressing water, waste, local production, soil health, and equitable distribution. Present with evidence from model simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Global Population, Agricultural Land Area
- **INTERNAL:** Crop Yield Per Hectare, Soil Health Index, Water Availability for Agriculture, Food Waste and Loss, Biodiversity in Agricultural Landscapes

Relationships

- **Global Population to Crop Yield Per Hectare:** POSITIVE (+) — Growing population increases demand pressure on agricultural production, pushing for higher yields per hectare through intensification of inputs and farming practices.
- **Crop Yield Per Hectare to Soil Health Index:** NEGATIVE (-) — Higher yields achieved through intensive chemical farming degrade soil biology, organic matter, and structure over time — the short-term production boost comes at the cost of long-term soil health.
- **Soil Health Index to Crop Yield Per Hectare:** POSITIVE (+) — Healthy soil with robust biology provides natural fertility, water retention, and pest suppression that support higher yields. As soil health declines, yields become increasingly dependent on external chemical inputs.
- **Water Availability to Crop Yield Per Hectare:** POSITIVE (+) — Adequate water supply is essential for crop growth. As water availability declines from aquifer depletion, yields drop even with sufficient soil nutrients and other inputs.
- **Biodiversity in Agricultural Landscapes to Crop Yield Per Hectare:** POSITIVE (+) — Pollinators, pest predators, and soil organisms provide ecosystem services essential for crop production. Biodiversity loss from pesticides and habitat destruction removes these natural supports.
- **Food Waste and Loss to Global Population:** NEGATIVE (-) — Higher food waste reduces the effective food supply relative to production, increasing the gap between what is grown and what is available for consumption, effectively raising the production requirement for each person fed.

Key Discoveries

- Industrial intensification produces short-term yield gains but long-term productivity collapse — the model shows Soil Health Index declining steadily under conventional practices until yields begin falling despite increasing chemical inputs
- Food Waste and Loss is the single largest leverage point in the food system — reducing waste by 50% is equivalent to increasing agricultural land by 25% without any environmental cost
- Water Availability becomes the binding constraint before land in most scenarios — aquifer depletion from irrigation is irreversible on human timescales and threatens food production in the world\'s most productive agricultural regions
- Sustainable intensification that combines waste reduction, soil regeneration, and biodiversity protection can theoretically feed 10 billion people on current agricultural land — but requires fundamental transformation of farming practices and food distribution systems

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We just need to grow more food to solve hunger"	Global food production already exceeds the caloric needs of 10 billion people, yet 800 million are hungry. Hunger is primarily a distribution and access problem, not a production problem. Poverty, conflict, infrastructure gaps, and political failures prevent food from reaching those who need it. Additionally, one-third of food produced is wasted. Simply growing more food within the current system would increase environmental damage without solving the underlying distribution failures.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Organic farming can\'t feed the world because yields are lower"	While organic yields are typically 10-20% lower than conventional in developed countries, this comparison ignores several critical factors: conventional yields are declining as soil degrades, organic soil health improves over time (closing the yield gap), the one-third of food wasted means production capacity far exceeds actual needs, and conventional farming\'s dependence on fossil fuel-based inputs and depleting aquifers makes it unsustainable long-term. The real question is not which system produces more per acre in year one, but which system still produces food in year fifty.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Technology will solve the food crisis without changing how we farm or eat"	While agricultural technology (precision farming, GMOs, vertical farms) can increase efficiency, technology cannot override the fundamental constraints of soil biology, water availability, and ecosystem services. The model shows that yield improvements are temporary if they come at the expense of the ecological foundation. Furthermore, the most impactful interventions — reducing waste, improving distribution, and shifting diets — are behavioral and systemic, not technological. Technology is a tool, not a solution in itself.